



IBM Skills Build for University Engagements

AICTE-2026

Problem Statements on Agentic AI

Problem Statement No.1 - Research Agent

The Challenge- A **Research Agent** is an AI system designed to assist with academic and scientific research tasks. It can autonomously search for literature, summarize papers, and organize references. Using natural language processing, it understands research questions and retrieves relevant information.

The agent can generate reports, suggest hypotheses, and even draft sections of research papers.

It saves time by automating repetitive tasks like citation management and data extraction.

Research Agents enhance efficiency, accuracy, and innovation in both academic and industrial R&D.

Technology - Use of IBM cloud lite services /IBM Granite is mandatory.

ResearchPilot

1. INTRODUCTION

The rapid growth of scientific and academic publications has made it increasingly difficult for researchers, students, and professionals to efficiently access, understand, and utilize relevant knowledge. Research papers often contain complex methodologies, technical terminology, and extensive experimental results that require significant time and effort to analyze.

Traditional literature review processes involve manually reading multiple research papers, extracting key insights, comparing methodologies, and identifying research gaps. This process is time-consuming and can limit research productivity.

Artificial Intelligence offers an opportunity to simplify research workflows by enabling intelligent document analysis, semantic search, and automated knowledge extraction. By combining document retrieval techniques with large language models, researchers can interact with academic content more efficiently and gain insights faster.

2. PROBLEM STATEMENT

Researchers frequently face challenges when working with large collections of research papers due to:

- Time-consuming manual reading and analysis
- Difficulty identifying important contributions and findings
- Challenges in comparing methodologies across papers
- Inefficient information retrieval from lengthy documents
- Limited tools for interactive exploration of academic content

There is a need for an intelligent system capable of understanding research documents and assisting users in extracting meaningful insights through natural language interaction.

3. PROJECT OBJECTIVE

The objective of ResearchPilot is to develop an AI-powered Research Agent that assists users in exploring, understanding, and analyzing research papers using IBM Watsonx and Granite Foundation Models.

The system aims to:

- Simplify research paper exploration
- Enable semantic search across academic documents
- Generate concise paper summaries

- Support question answering from research content
- Extract methodologies and key findings
- Provide evidence-based responses grounded in source documents
- Improve efficiency in literature review and knowledge discovery

4. PROPOSED SOLUTION

ResearchPilot is an AI-powered Research Agent built using IBM Watsonx.ai, IBM Granite Foundation Models, Retrieval-Augmented Generation (RAG), FAISS, and Flask.

The application allows users to upload research papers in PDF format and automatically process their content. Documents are converted into vector embeddings and stored within a FAISS vector database.

When a user submits a question, the system retrieves the most relevant document sections using semantic similarity search. The retrieved context is then provided to the IBM Granite model through IBM Watsonx.ai to generate contextual and document-grounded responses.

This approach combines information retrieval with language generation to create an intelligent research assistant capable of supporting academic research activities.

5. SYSTEM WORKFLOW

- Step 1:** Upload research papers in PDF format.
- Step 2:** Extract text and metadata from documents.
- Step 3:** Split documents into meaningful chunks.
- Step 4:** Generate vector embeddings.
- Step 5:** Store embeddings in FAISS vector database.
- Step 6:** User submits a research query.
- Step 7:** Retrieve relevant document sections using semantic search.
- Step 8:** Provide retrieved context to IBM Granite model.
- Step 9:** Generate contextual response.
- Step 10:** Display answer along with supporting evidence and source references.

6. TECHNOLOGY STACK

IBM Watsonx.ai AI development platform used for model access, Prompt Lab experimentation, and document-grounded AI workflows.	IBM Granite Foundation Models Used for contextual response generation, summarization, and question answering.
Retrieval-Augmented Generation (RAG) Combines retrieval and generation to improve answer relevance and grounding.	FAISS Vector Database Provides efficient semantic similarity search across research documents.
Python Core programming language used for backend development.	Flask Lightweight web framework used to build the ResearchPilot application.
HTML, CSS, JavaScript Used for frontend user interface development.	HuggingFace Embeddings Used to convert document chunks into vector representations.

7. KEY FEATURES

- Research Paper Upload and Processing
- Research Paper Summarization
- Methodology Extraction
- Evidence-Based Responses
- Semantic Search and Information Retrieval
- Question Answering System
- Multi-Document Analysis
- User-Friendly Web Interface

8. EXPECTED OUTCOMES

The proposed solution will enable researchers and students to interact with research papers more efficiently. Users will be able to retrieve relevant information, generate summaries, explore methodologies, and gain insights through natural language interaction.

The system reduces the effort required for manual literature review and demonstrates the practical application of IBM Watsonx and Granite Foundation Models in academic research assistance.

9. CONCLUSION

ResearchPilot demonstrates how Retrieval-Augmented Generation and IBM Granite Foundation Models can be applied to academic research workflows. By combining semantic retrieval, document grounding, and AI-powered response generation, the system provides an intelligent platform for research paper exploration and knowledge discovery.